

### 20.3 Discussion Questions

#### ❖ The Nuclear Atom

- D1 (a) In the  $\alpha$ -particle scattering experiment,  $\alpha$ -particles, travelling in a vacuum, are incident on a gold foil. Sketch diagrams to illustrate the path of an  $\alpha$ -particle if its original path
- (i) is directly towards the nucleus of a gold atom,
  - (ii) passes close to the nucleus of a gold atom,
  - (iii) passes some distance from the nucleus.
- (b) Describe and explain how the  $\alpha$ -particle scattering experiment which you have illustrated in part (a) gives evidence for the existence and small size of the nucleus.

- D2 During his discovery of the neutron in 1932, James Chadwick, determined the mass of the newly identified particle by firing a beam of fast neutrons towards known targets. As the speed of the neutrons cannot be determined by studying its motion within an electric or a magnetic field, he directed a beam of neutrons (all having the same speed), at two different targets and measuring the maximum recoil speeds of the target nuclei.
- (a) Explain why the speed of neutrons could not be determined by studying its motion within an electric or a magnetic field.
  - (b) Explain the origin of the maximum recoil speed of a target nucleus.
  - (c) Let the masses of the two target nuclei be  $m_1$  and  $m_2$  and the final speeds of the target nuclei be  $v_1$  and  $v_2$ , respectively.

Show that the mass of the neutron can be calculated from the equation

$$m_n = \frac{m_1 v_1 - m_2 v_2}{v_2 - v_1}$$

- (d) Use the following data to calculate the mass of the neutron.

$$m_p = 1.00 \text{ u}; m_N = 14.00 \text{ u}, v_p = 3.30 \times 10^7 \text{ m s}^{-1}, v_N = 4.70 \times 10^6 \text{ m s}^{-1}$$

Masses of a proton and nitrogen nuclei are taken to be 1.00 u and 14.0 u, respectively,

#### ❖ Radioactivity

- D3 (a) The radioactive decay of the isotope cobalt-60 (beta emitter) is both spontaneous and random. Explain what is meant by
- (i) spontaneous decay,
  - (ii) random decay.
- (b) A student has a sample of cobalt-60 – a beta emitter – inside a box. The student defines the half-life of cobalt-60 as the time taken for the number of nuclei inside the box to decay to one half of its initial value.

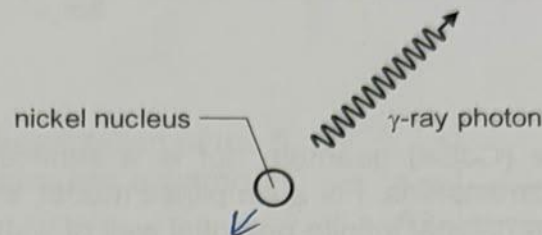
State and explain one reason why this definition is inappropriate.

[Modified N09/3/4]

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D3 A stationary nickel nucleus of mass  $9.95 \times 10^{-26}$  kg emits a  $\gamma$ -ray photon of energy 1.17 MeV.

- (a) For the  $\gamma$ -ray photon,
- (i) show that its wavelength is approximately  $1.06 \times 10^{-12}$  m,
  - (ii) calculate its momentum.
- (b) The direction in which the  $\gamma$ -ray photon is emitted is as shown.



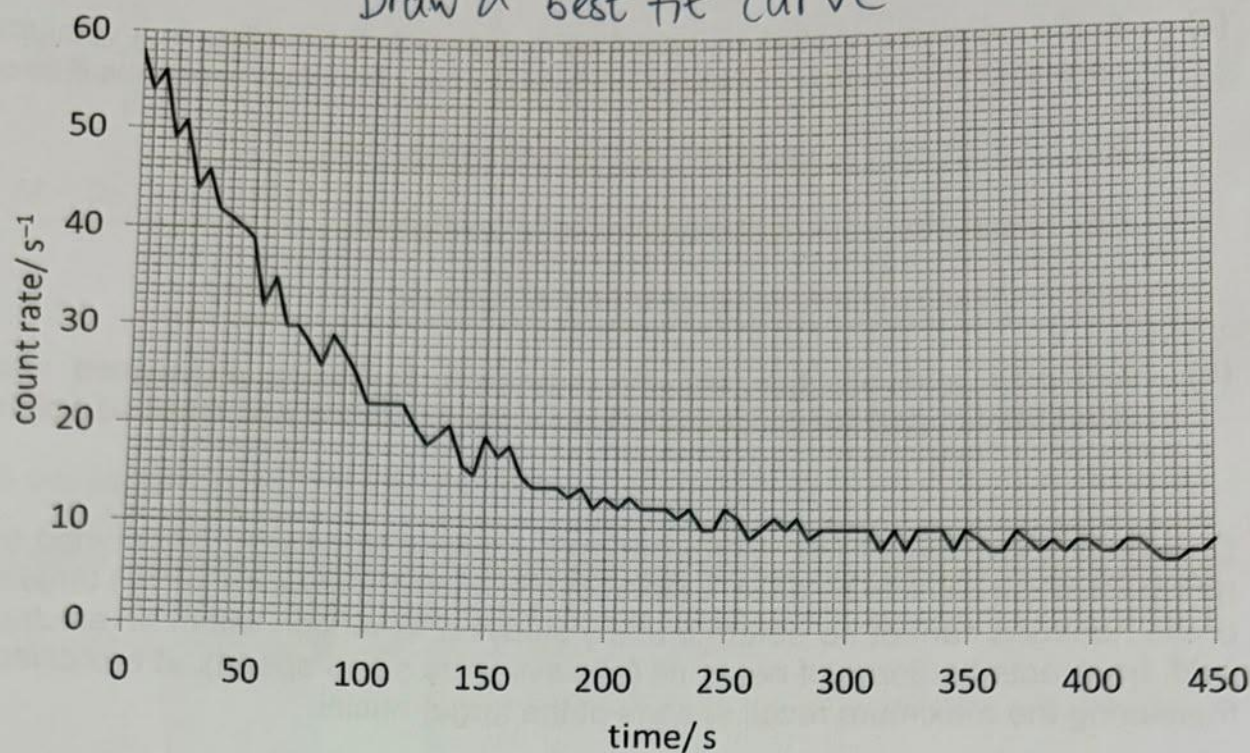
- (i) On the figure, draw an arrow to show the direction of motion of the nickel nucleus after the emission of the photon.
  - (ii) Calculate the speed of the nickel nucleus after the emission of the photon.
- (c) A second nickel nucleus that is moving emits a  $\gamma$ -ray photon. Suggest, with a reason, whether angle between the final direction of motion of the nucleus and that of the emitted photon will be the same as that in the figure.



D4

An experiment is carried out in which the count rate is measured at a fixed distance from a sample of a certain radioactive material. The figure shows the variation of count rate with time that was obtained.

Draw a best fit curve



- (a) Use the graph to estimate the half-life of the material.
- (b) Explain how you would minimise the uncertainty in determining the half-life of the material due to
  - (i) the random nature of the count rate,
  - (ii) the background radiation.

[Modified N91/2/7]

D5

The hospital treatment of internal tumours requires the use of 2.0 mg of the radionuclide  $^{60}_{27}\text{Co}$ . The half-life of  $^{60}_{27}\text{Co}$  is 5.3 years and each decay emits an electron of average energy 0.12 MeV followed by  $\gamma$  rays of total energy 2.5 MeV.

- (a)
  - (i) What is meant by the term "decay constant"?
  - (ii) Determine the decay constant of  $^{60}_{27}\text{Co}$ .
- (b) Calculate
  - (i) the number of nuclei in 2.0 mg of  $^{60}_{27}\text{Co}$ ,
  - (ii) the initial activity of  $^{60}_{27}\text{Co}$ ,
  - (iii) the initial power output from 2.0 mg of  $^{60}_{27}\text{Co}$ ,
  - (iv) the power output from 2.0 mg of  $^{60}_{27}\text{Co}$  after 21.2 years.



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- (c) Explain why the energy absorbed by the internal organs in the body is smaller than the energy emitted by the source.
- (d) Nowadays, linear accelerators emitting "megavolt" x-rays are used instead of  ${}^{60}_{27}\text{Co}$ . Suggest two reasons why linear accelerators may be preferred.

[N88/2/12]

**D6** A geologist tries to determine the age of a sample of rock containing  ${}^{40}_{19}\text{K}$  which decays to give a stable nuclide  ${}^{40}_{18}\text{Ar}$ . The activity of the sample is found to be 2.4 Bq, while the original activity of a similar rock having the same mass is 38.4 Bq. The decay constant of  ${}^{40}_{19}\text{K}$  is  $5.33 \times 10^{-10} \text{ yr}^{-1}$ .

- (a) (i) Define half-life.
- (ii) Determine the half-life of  ${}^{40}_{19}\text{K}$ .
- (iii) Determine the age of the sample of rock.
- (b) (i) Determine the number of  ${}^{40}_{19}\text{K}$  nuclei when the activity is 2.4 Bq.
- (ii) State whether the half-life of  ${}^{40}_{19}\text{K}$  will change if
1. a sample of greater mass of  ${}^{40}_{19}\text{K}$  is chosen, and
  2. the temperature of the sample of  ${}^{40}_{19}\text{K}$  is increased.

❖ **Nuclear Reactions**

**D7** When a deuteron (nucleus of deuterium) of mass 2.013553 u and negligible kinetic energy fuses with a Lithium-6 nucleus of mass 6.013476 u, the compound nucleus disintegrates spontaneously into two  $\alpha$ -particles, each of mass 4.001505 u.

Calculate the energy, in joules, of each  $\alpha$ -particle.

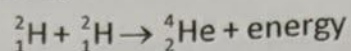
**D8** When  ${}^{235}_{92}\text{U}$  is bombarded by slow neutrons,  ${}^{141}_{56}\text{Ba}$  and  ${}^{92}_{36}\text{Kr}$  may be produced.

- (a) Write the equation for this nuclear reaction.
- (b) Name this type of nuclear reaction.
- (c) If the binding energy per nucleon of  ${}^{235}_{92}\text{U}$  is 7.6 MeV and that of both  ${}^{141}_{56}\text{Ba}$  and  ${}^{92}_{36}\text{Kr}$  is 8.5 MeV, show that 195 MeV of energy is released in one such reaction.
- (d) Suggest two forms of energy which the energy released in (c) is transformed into.

[Modified J97/2/7]



- D9 The nuclear fusion of deuterium-2 ( ${}^2_1\text{H}$ ) to form helium-4 is proposed to produce electrical power. The nuclear reaction is given by



The atomic masses of deuterium-2 and helium-4 are  $2.014102\text{ u}$  and  $4.002603\text{ u}$  respectively.

- (a) Calculate the decrease in mass that occurs in the reaction.
- (b) Calculate the energy released from  $1.0\text{ g}$  of deuterium.
- (c) What is the mass of deuterium needed each day if the required power output of the fusion power plant is  $100\text{ MW}$  and the efficiency of electricity generation is  $40\%$ ?

#### 20.4 Numerical Answers

- D2  $1.16\text{ u}$
- D4  $59.2\text{ s}$
- D5  $0.13\text{ yr}^{-1}$  or  $4.14 \times 10^{-9}\text{ s}^{-1}$ ,  $2.0 \times 10^{19}$ ,  $8.3 \times 10^{10}\text{ Bq}$ ,  $0.035\text{ W}$ ,  $0.0022\text{ W}$
- D6  $1.30 \times 10^9\text{ yr}$ ,  $5.20 \times 10^9\text{ yr}$ ,  $1.42 \times 10^{17}$
- D7  $1.79 \times 10^{-12}\text{ J}$
- D9  $0.025601\text{ u}$ ,  $5.72 \times 10^{11}\text{ J}$ ,  $37.8\text{ g}$

#### 20.5 Self-Check & Self-Practice Suggested Solutions

- S1 The **positive charge** and **most of the mass** of the atom was **concentrated** in a **very small volume** called the nucleus and the electrons orbit around the nucleus like planets around the sun.
- S2 Mass number  $A$  is the **number of protons and neutrons** while atomic number  $Z$  is the **number of protons**.
- S3 A nuclide is an **atomic species** with specific mass number  $A$  and atomic number  $Z$ . Isotopes are nuclides with the same  $Z$  but differing  $A$  (or  $N$ ).  
Isotope is a subset of nuclide though both can be used interchangeably.
- S4 Radioactive decay is spontaneous – the **probability** of radioactive decay of an atom is **unaffected by external factors** such as temperature, pressure or chemical composition.  
Radioactive decay is also random – it is **impossible to predict when a particular atom will decay**.